Product Requirements Document (PRD): AI-Driven Fleet Management & Route Optimization Module

1. Product Overview & Vision

The goal of this module is to upgrade the existing "GREENIE Smart C&D Waste Monitoring System" by integrating rigorous Artificial Intelligence (AI) and Machine Learning (ML) models. While the current platform successfully provides real-time geospatial tracking, fleet registries, and operational dashboards, introducing AI will transition the system from a passive tracking tool into an active, predictive, and autonomous logistics engine. The focus will be on dynamic route optimization, load matching, and predictive maintenance.

2. Baseline System Analysis (Based on Current UI)

The AI integration will build upon the existing modules visible in the system:

- **Operations Dashboard:** Currently tracks 124 total units (84 active, 22 idle, 17 in maintenance) and pending collections (e.g., 23 awaiting dispatch).
- **Collection Sites:** Manages node capacities (e.g., CS-001 Chennai Central Hub at 50 tons, CS-002 Coimbatore North Depot at 80 tons).
- **Monitoring / Map:** Provides static live-feed plotting of active, idle, and maintenance sites across South India.
- **Vehicles Registry:** Logs specific vehicle data including plate numbers, models (Tata 407, Ashok Leyland Dost, BharatBenz 1617R), current mileage, and last service dates.
- **Analytics & Reports:** Tracks revenue (e.g., ₹659K), average collections per site/day (14.2), and waste by type.

3. Proposed AI Features & Functional Requirements

Feature 1: Dynamic Route Optimization & Dispatch (AI-Routing)

- **Requirement:** The system must autonomously assign the pending collections (currently shown as 23 on the dashboard) to the 22 idle units using AI-driven dispatching.
- **AI Model:** Implement Deep Q-Networks (DQN) and Reinforcement Learning (RL) models to solve the Capacitated Vehicle Routing Problem (CVRP).
- **Functionality:**
    - The algorithm must dynamically respond to real-time traffic conditions, weather forecasts, and road closures, recalculating the optimal path continuously.
    - It must perform "eco-routing" to minimize harsh accelerations and gradients, thereby reducing fuel burn and carbon emissions.
- **UI Integration:** The *Monitoring / Map* module must be updated to display AI-generated dynamic route lines (via Mapbox/OpenStreetMap integration) rather than just point-in-time vehicle locations.

Feature 2: AI-Assisted Load & Capacity Matching

- **Requirement:** Maximize the utilization of the fleet based on specific vehicle capacities and waste types.
- **AI Model:** Utilize Fractional Knapsack dynamic programming paired with Deep Q-Networks to maximize vehicle capacity utilization up to 100%.
- **Functionality:** The AI must match heavy payloads (e.g., concrete and rubble) with heavy trucks like the Mahindra Furio or BharatBenz 1617R (from the *Vehicles* registry), while routing lighter trucks like the Tata 407 for smaller, mixed-debris pickups.

Feature 3: Predictive Fleet Maintenance

- **Requirement:** Reduce the number of vehicles in maintenance (currently at 17, representing a significant portion of the 124 total units) through preemptive servicing.
- **AI Model:** Deploy component-level health models using historical telematics and ML algorithms.
- **Functionality:**
    - Analyze real-time vibration, temperature, and voltage data alongside the existing mileage and "last service" dates in the *Vehicles* registry.
    - Flag anomalies and predict component failures before roadside breakdowns occur.
- **UI Integration:** Add an "AI Health Score" column to the *Vehicles* table and generate automated alerts in the *Operations Dashboard* for prescriptive slotting (booking maintenance during route gaps to preserve utilization).

4. Data & Technical Requirements

- **Data Inputs:** Real-time GPS telematics, traffic data APIs (e.g., Google Maps or SUMO for traffic simulation), vehicle specifications, and load cell weights from the transfer stations.
- **Backend Architecture:** A scalable Python/Flask backend to run the CVRP algorithms and process the RL models without lagging the frontend dashboard.
- **Hardware Integration:** Ensure compatibility with existing IoT sensors on the trucks and the transfer station's weighbridges to feed accurate payload data into the routing algorithm.

5. Success Metrics (KPIs)

To prove the viability and profitability of this AI integration to investors, the system will target the following metrics:

- **Distance & Fuel Reduction:** Achieve a 25% to 30% reduction in total travel distance and fuel consumption compared to baseline heuristic routing methods.
- **Fleet Utilization:** Increase total vehicle capacity utilization to near 100%, allowing the same volume of waste (e.g., 355 T) to be collected with up to 20% fewer deployed vehicles.
- **Downtime Reduction:** Decrease the percentage of the fleet "In Maintenance" (currently at ~13%) by at least half through predictive rather than reactive repairs.